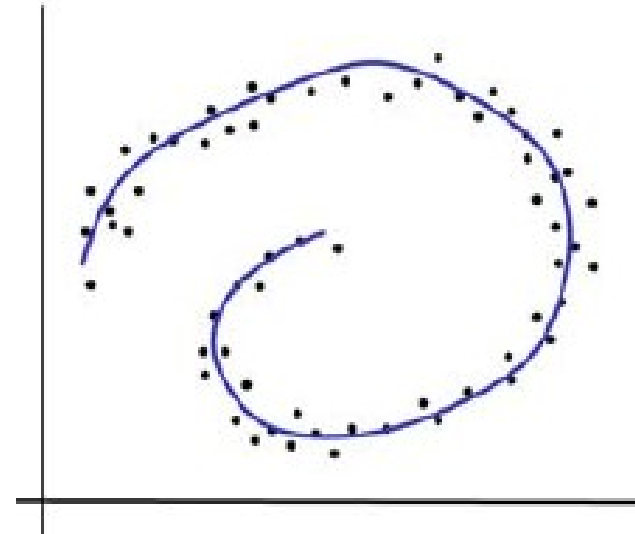
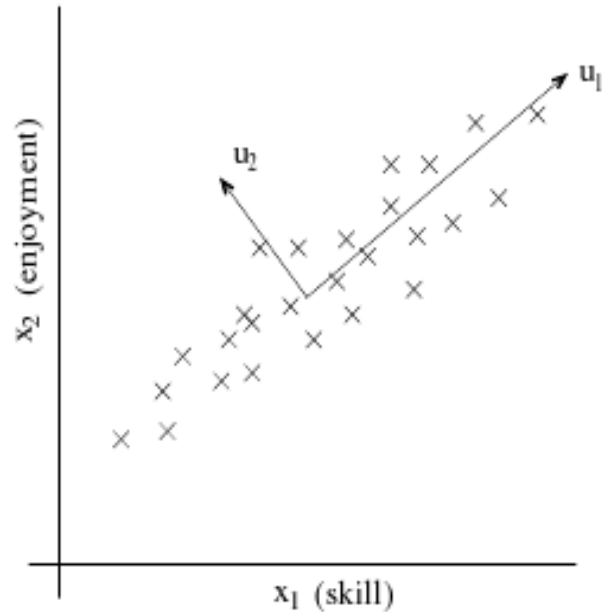


Foundations of Artificial Intelligence

Feature Selection

Andrea Torsello

- We have seen that high dimensional data is hard to classify (curse of dimensionality)
- Often however, the data does not fill all the space
- Rather, it lies (approximately) on a lower dimensional manifold (surface)

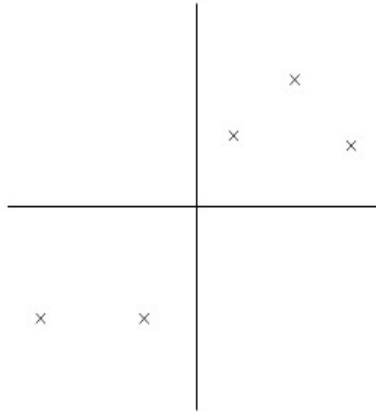


- Finding this manifold means finding a low-dimensional parametrization that captures the essence of the data (small error from data-point to parametrized point on manifold)
- Principal Component Analysis (PCA) assumes that the data lies on a linear subspace and helps us find the subspace

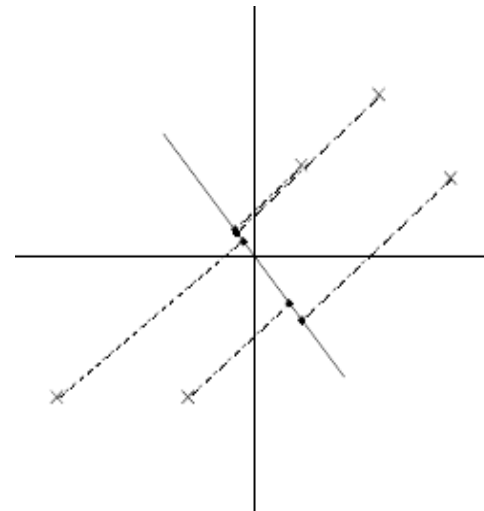
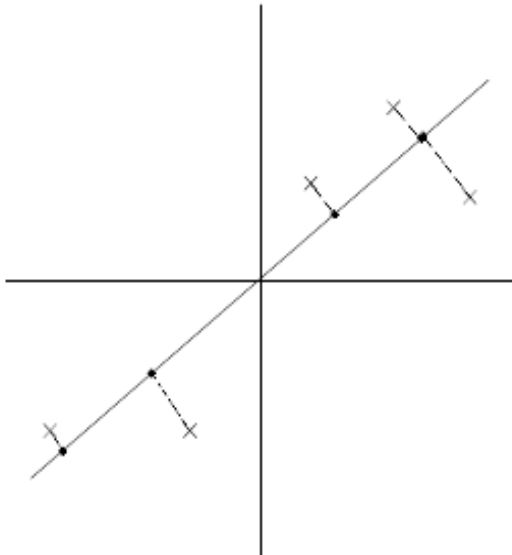
PCA

- There are two common definitions used for PCA that give rise to the same algorithm.
 - PCA is the orthogonal projection of the data onto a linear subspace (principal subspace) such that the variance of the projected data is maximized
 - PCA is the projection onto a linear subspace that minimizes the mean squared distance of the data points from their projections

- Consider the following dataset...



- And the following projection to linear subspaces



Maximum variance formulation

- Let $\mathbf{u}$ be a unit vector (i.e., $\mathbf{u}^T \mathbf{u} = 1$)
 - The mean of the data along $\mathbf{u}$ is $\mathbf{u}^T \bar{\mathbf{x}}$ where $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$
 - The variance of the projected data is

$$\frac{1}{n} \sum_{i=1}^n (\mathbf{u}^T \mathbf{x}_i - \mathbf{u}^T \bar{\mathbf{x}})^2 = \mathbf{u}^T \mathbf{S} \mathbf{u}$$

- where $\mathbf{S}$ is the covariance matrix

$$\mathbf{S} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$$

- Thus the variance is maximized by the unit vector $\mathbf{u}$ that maximizes $\mathbf{u}^T \mathbf{S} \mathbf{u}$
 - Leading eigenvector!
- This eigenvector is known as the principal component
- We can define additional principal components incrementally
 - Choose a new direction $\mathbf{u}$ that maximizes the variance among the vectors orthogonal to the directions already considered
- In general the k principal components correspond to the k leading eigenvectors

Reconstruction and error

- Let $\{\mathbf{u}_i\}_{i=1,\dots,k}$ be a set of principal components
- Each data point can be approximated by a linear combination of the components

$$\mathbf{x}_i - \bar{\mathbf{x}} \approx \sum_{j=1}^k \alpha_{ij} \mathbf{u}_j \qquad \mathbf{x}_i \approx \bar{\mathbf{x}} + \sum_{j=1}^k \alpha_{ij} \mathbf{u}_j$$

- Since the basis is orthonormal, we can obtain the coordinates by orthogonal projection

$$\alpha_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^T \mathbf{u}_j$$

- Thus the vector $\alpha_i = (\alpha_{i1}, \dots, \alpha_{ik})^T$ is a parametrization of a point in the k-dimensional principal subspace
- But how far is the actual point from its projection onto the principal component?
- On a D-dimensional principal subspace (the whole space) the reconstruction would be perfect.
- By limiting ourselves to the first k principal components, we have

$$\mathbf{x}_i - \tilde{\mathbf{x}}_i = \sum_{j=k+1}^D ((\mathbf{x}_i - \bar{\mathbf{x}})^T \mathbf{u}_j) \mathbf{u}_j$$

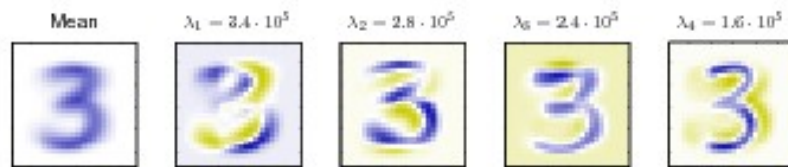
- The *average squared distance* is

$$\frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - \tilde{\mathbf{x}}_i\|^2 = \frac{1}{n} \sum_{i=1}^n \sum_{j=k+1}^D ((\mathbf{x}_i - \bar{\mathbf{x}})^T \mathbf{u}_j)^2 = \sum_{j=k+1}^D \mathbf{u}_j^T \mathbf{S} \mathbf{u}_j$$

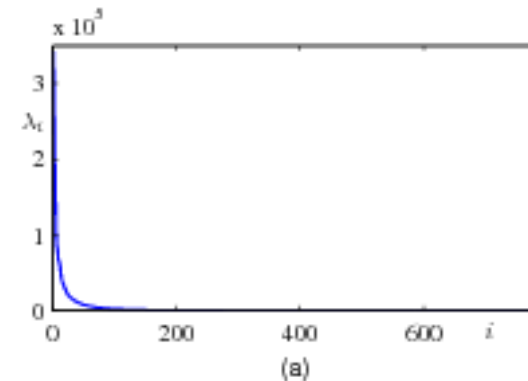
- Which is minimized when the remaining D-k components are associated to the smallest eigenvalues

Applications of PCA

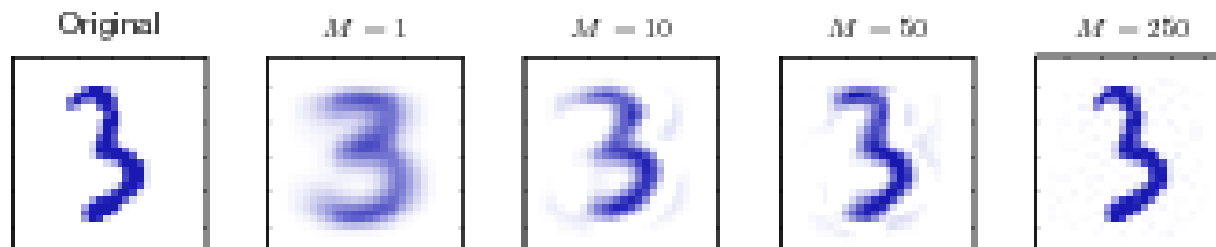
- PCA is used when the dimensionality of the problem is huge and there is a lot of redundancy
- This is typically the case in image analysis tasks



Mean vector and first four eigenvectors for the digit dataset



Eigenvectors of the digit dataset



Reconstruction using 1, 10, 50, and 250 components

PCA and Normalization

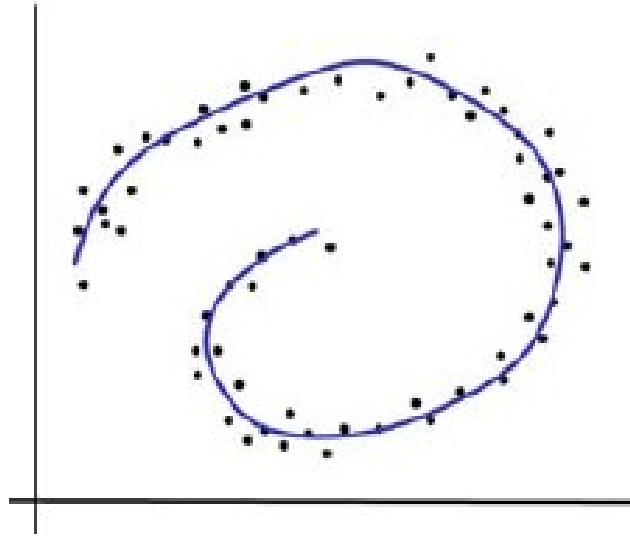
- When talking about distances we referred to the problem of putting the features on a similar scale.
- One approach suggested was to *standardize* the data, i.e. scale it so that each feature had zero mean and unit variance
- However, standardized data can still be correlated (thin diagonal axis of the ellipsoid)
- PCA allows us to operate a stronger normalization: it allows us to transform the data so that it has zero mean and identity covariance matrix.
- Let $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$ where
 - $\mathbf{S}$ is the data covariance matrix;
 - $\mathbf{U}$ is an orthogonal matrix composed of the eigenvectors of $\mathbf{S}$;
 - $\mathbf{\Lambda}$ is a diagonal matrix containing the eigenvalues of $\mathbf{S}$.
- We can transform the data mapping each point $\mathbf{x}_i$ onto $\mathbf{y}_i = \mathbf{\Lambda}^{-1/2}\mathbf{U}^T(\mathbf{x}_i - \bar{\mathbf{x}})$
- The new data has clearly zero mean, and has identity covariance, in fact

$$\frac{1}{n} \sum_{i=1}^n \mathbf{y}_i \mathbf{y}_i^T = \frac{1}{n} \sum_{i=1}^n \mathbf{\Lambda}^{-1/2} \mathbf{U}^T (\mathbf{x}_i - \bar{\mathbf{x}}) (\mathbf{x}_i - \bar{\mathbf{x}})^T \mathbf{U} \mathbf{\Lambda}^{-1/2} =$$
$$\mathbf{\Lambda}^{-1/2} \mathbf{U}^T \mathbf{S} \mathbf{U} \mathbf{\Lambda}^{-1/2} = \mathbf{\Lambda}^{-1/2} \mathbf{\Lambda} \mathbf{\Lambda}^{-1/2} = \mathbf{I}$$

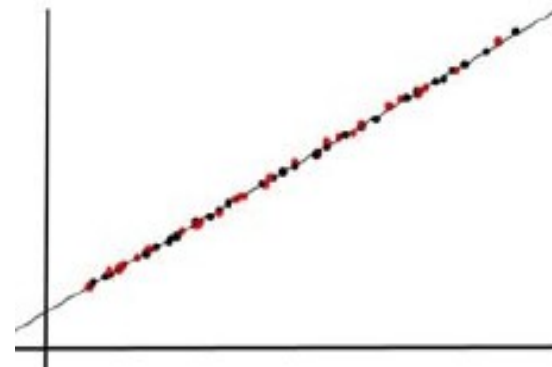
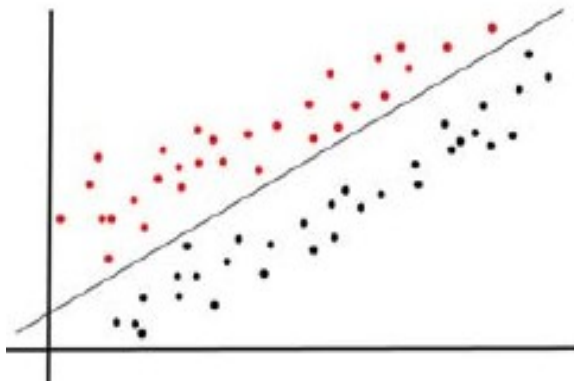
- This process is called *Whitening* or *Sphereing*

Limits of PCA

- Finds only linear subspaces, but the data can be on a more complex manifold

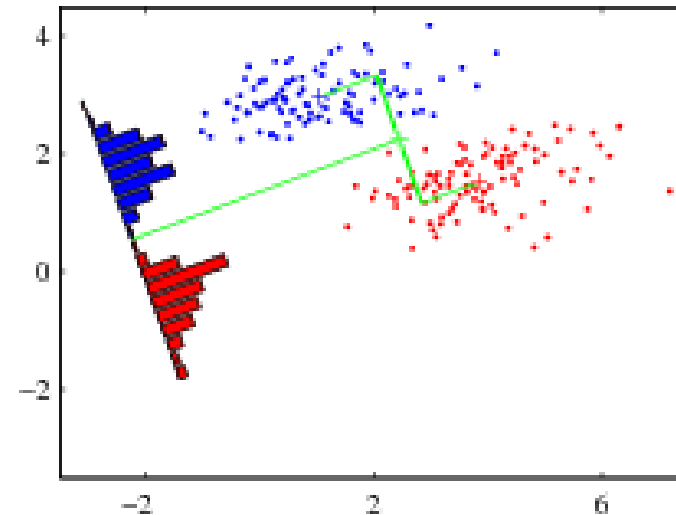
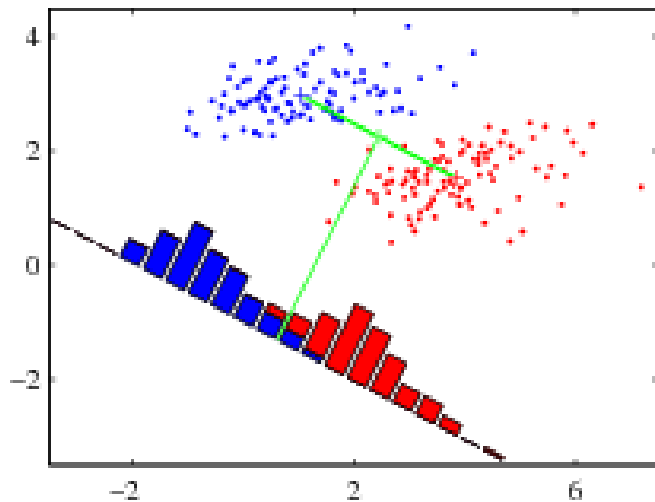


- It is insensitive to the classification task



Fisher discriminant analysis

- Fisher's linear discriminant tries to project the data on the one-dimensional subspace that maximizes the class discriminability



- We transform the data using $y = \mathbf{w}^T \mathbf{x}$.
- Let m_k be the mean of class k , the within class variance is $\sigma_k^2 = \sum_{i \in \mathcal{C}_k} (y_i - m_k)^2$

- The fisher criterion is $J(\mathbf{w}) = \frac{(m_2 - m_1)^2}{\sigma_1^2 + \sigma_2^2} = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}$

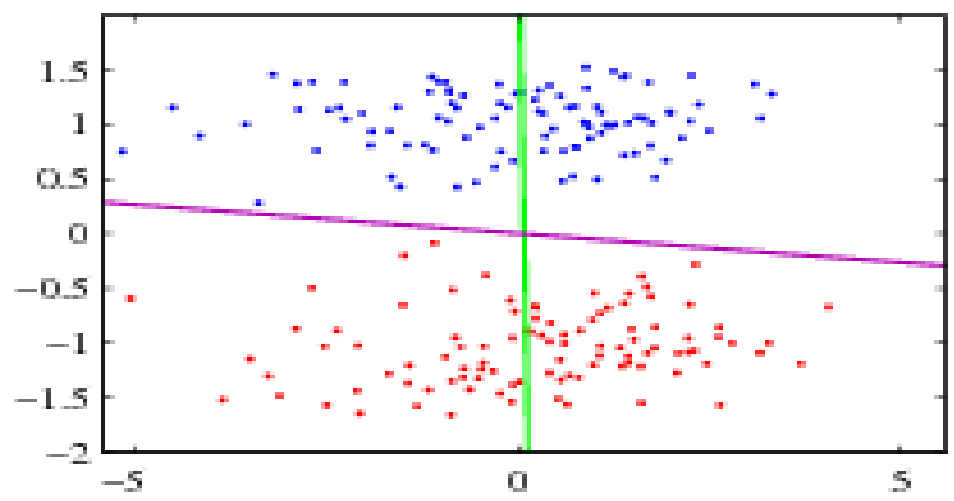
with

- Between class covariance matrix $\mathbf{S}_B = (\mathbf{m}_2 - \mathbf{m}_1)(\mathbf{m}_2 - \mathbf{m}_1)^T$
- Within class covariance matrix $\mathbf{S}_W = \sum_{i \in \mathcal{C}_1} (\mathbf{x}_i - \mathbf{m}_1)(\mathbf{x}_i - \mathbf{m}_1)^T + \sum_{i \in \mathcal{C}_2} (\mathbf{x}_i - \mathbf{m}_2)(\mathbf{x}_i - \mathbf{m}_2)^T$

- $J(\mathbf{w})$ is minimized when

$$(\mathbf{w}^T \mathbf{S}_B \mathbf{w}) \mathbf{S}_W \mathbf{w} = (\mathbf{w}^T \mathbf{S}_W \mathbf{w}) \mathbf{S}_B \mathbf{w}$$

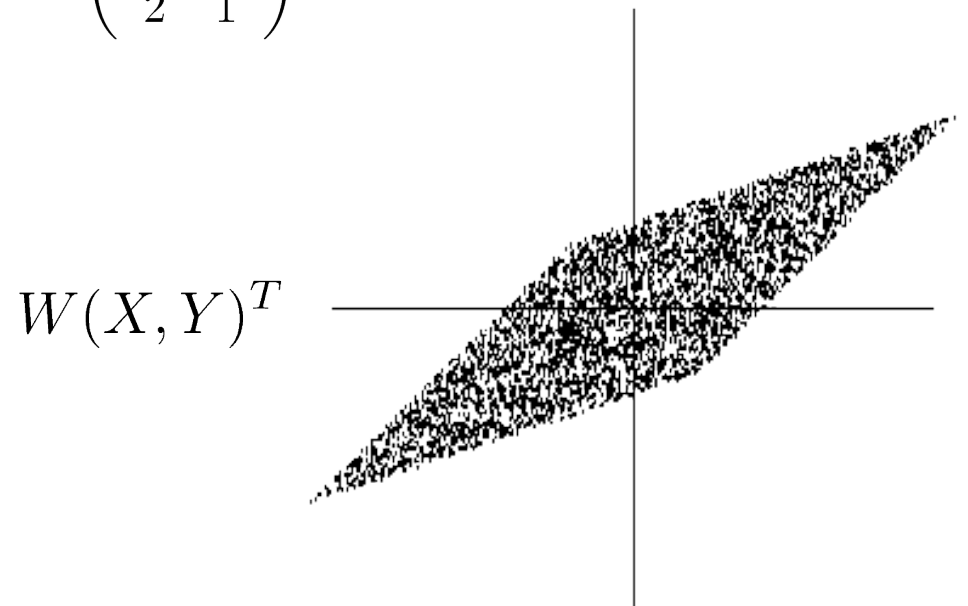
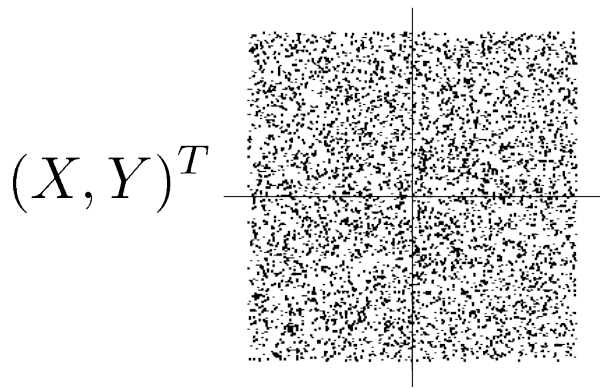
or $\mathbf{w} \propto \mathbf{S}_W^{-1}(\mathbf{m}_2 - \mathbf{m}_1)$



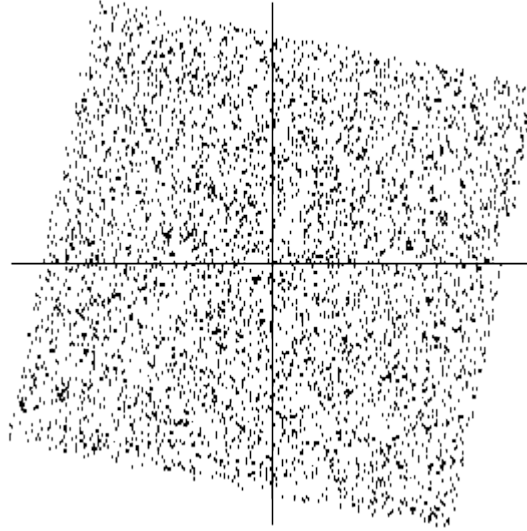
difference between principal component (purple) and Fisher discriminant (green) on the whitened Old Faithful dataset.

Independent component Analysis

- Principal Component Analysis provides a new orthogonal basis on which data is decorrelated
 - Whitening
- Is decorrelation enough?
 - Not necessarily!
- We would want each dimension to give “orthogonal” information
 - Decorrelation does not imply independence
- Let us assume X Y independent random variables uniform on $[-1; 1]$
- Let us mix them through a linear function $W = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$



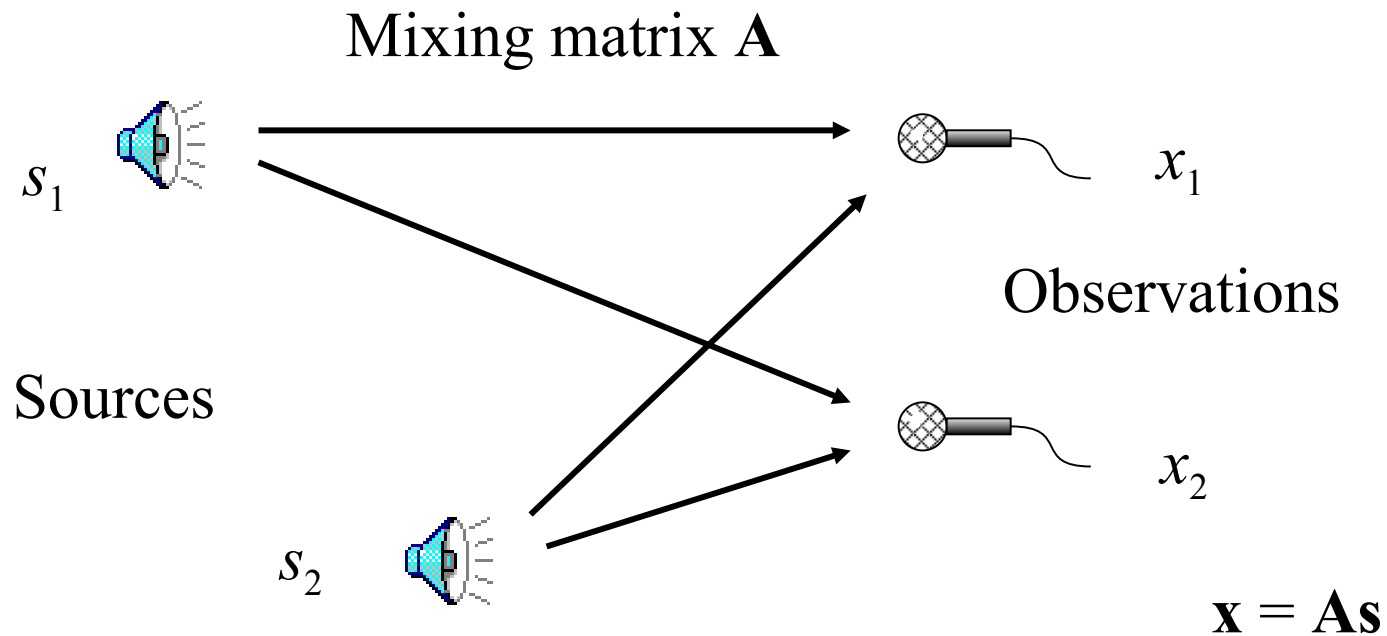
- If we perform whitening we obtain the following distribution



- The two variables are not independent!

- “Independent component analysis (ICA) is a method for finding underlying factors or components from multivariate (multi-dimensional) statistical data. What distinguishes ICA from other methods is that it looks for components that are both statistically *independent*, and *nonGaussian*.”
- Independent Component Analysis (ICA) is the identification & separation of mixtures of sources with little prior information.
- While PCA seeks directions that represents data best in a $\sum |\mathbf{x}_0 - \mathbf{x}|^2$ sense, ICA seeks such directions that are most independent from each other.
- Let $x_1(t), x_2(t) \dots x_n(t)$ be a set of observations of random variables
 - where t is the time or sample index
- Assume observation of the linear mixture
 - $\mathbf{y} = \mathbf{W}\mathbf{x}$ ($\mathbf{W}$ is unknown)
- ICA consists of estimating $\mathbf{W}$ and $\mathbf{x}$ from $\mathbf{y}$

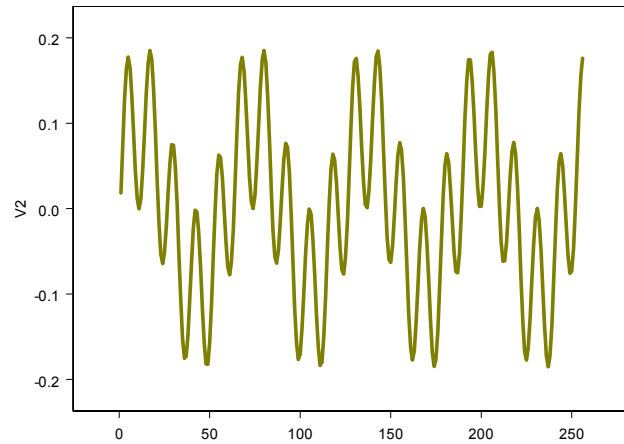
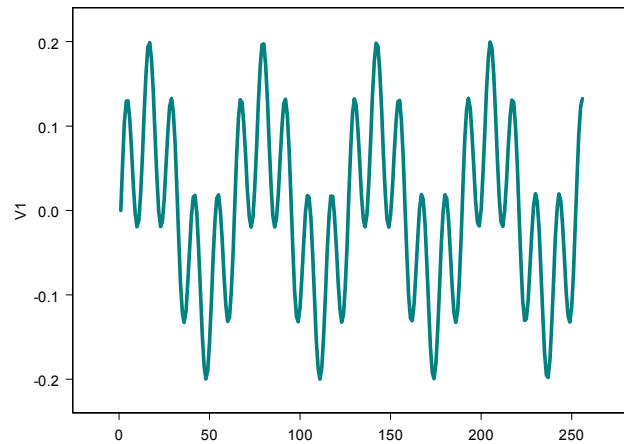
The simple “Cocktail Party” Problem



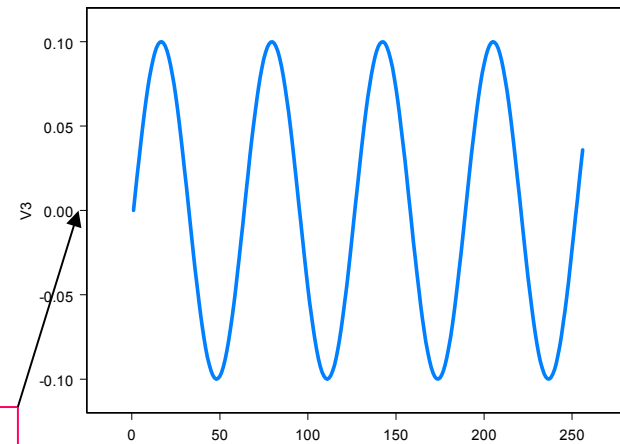
n sources, $m=n$ observations

Classical ICA (fast ICA) estimation

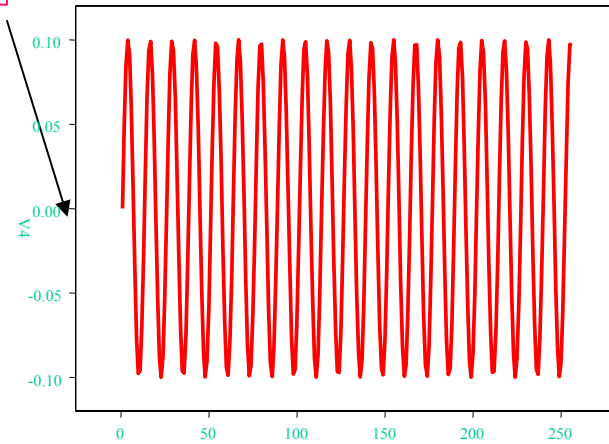
Observing signals

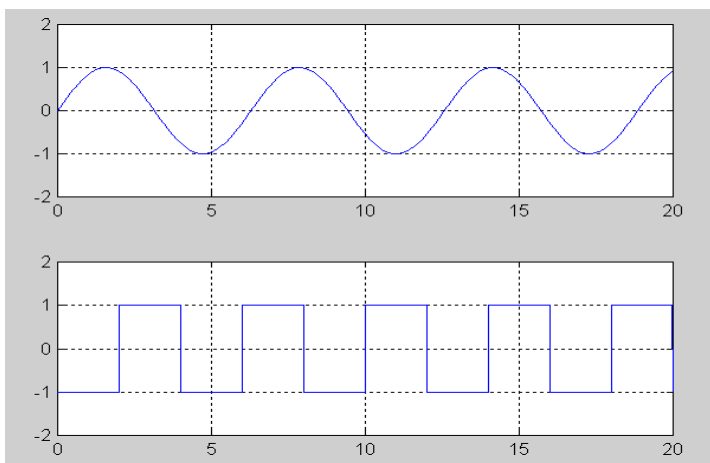


Original source signal

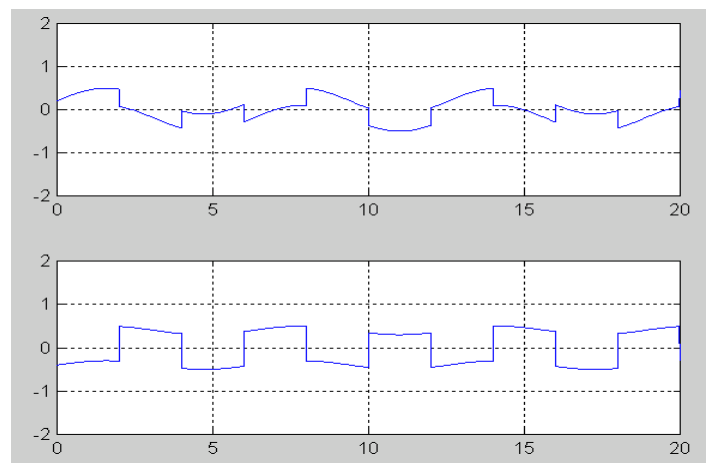


ICA

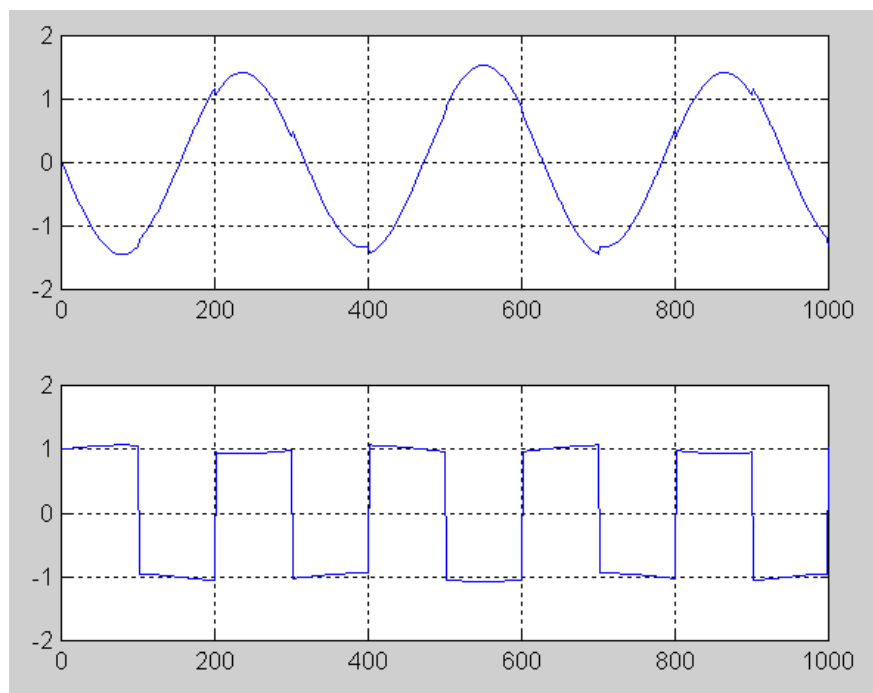




Two Independent Sources



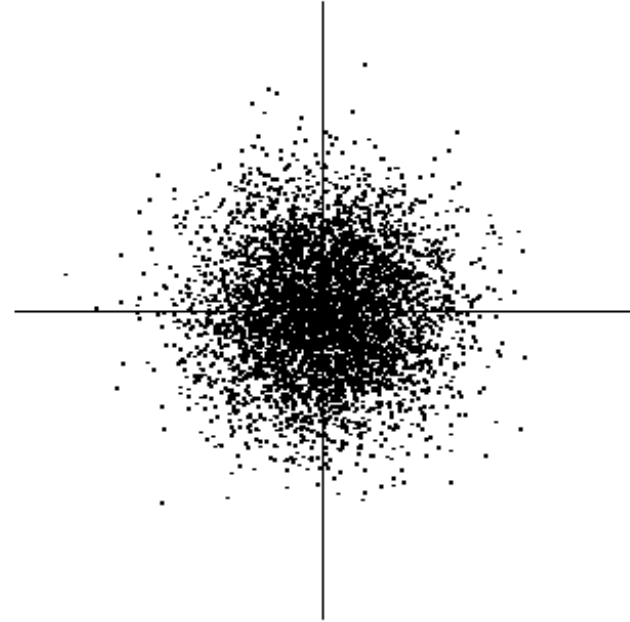
Mixture at two Mics



Get the Independent Signals out of the Mixture

Restrictions

- s_i are statistically independent
 - $p(s_1, s_2) = p(s_1)p(s_2)$
- Nongaussian distributions
 - The joint density of unit variance s_1 & s_2 is symmetric. So it doesn't contain any information about the directions of the cols of the mixing matrix A . So A can't be estimated.
 - If only one IC is gaussian, the estimation is still possible.

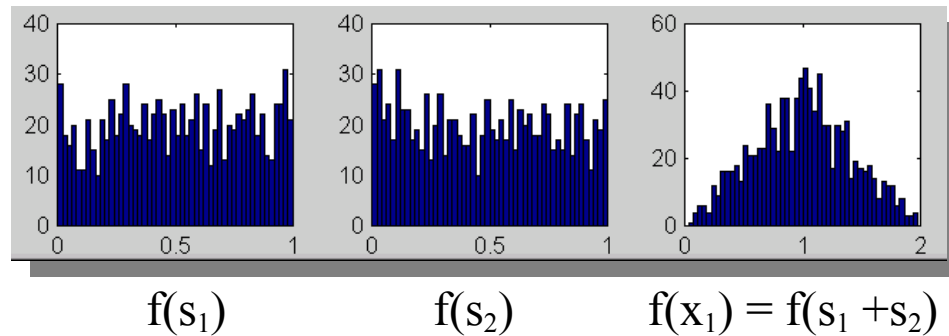


Ambiguities

- Can't determine the variances (energies) of the IC's
 - Both $\mathbf{s}$ & $\mathbf{A}$ are unknowns, any scalar multiple in one of the sources can always be cancelled by dividing the corresponding col of $\mathbf{A}$ by it.
 - Fix magnitudes of IC's assuming unit variance: $E\{s_i^2\} = 1$
 - Only ambiguity of sign remains
- Can't determine the order of the IC's
 - Terms can be freely changed, because both $\mathbf{s}$ and $\mathbf{A}$ are unknown. So we can call any IC as the first one.
- Can't reduce the dimensionality!

ICA Principle (Non-Gaussian is Independent)

- Key to estimating A is non-gaussianity
- The distribution of a sum of independent random variables tends toward a Gaussian distribution. (By CLT)



- Where $\mathbf{w}$ is one of the rows of matrix $\mathbf{W}$.

$$\mathbf{y} = \mathbf{w}^T \mathbf{x} = \mathbf{w}^T \mathbf{A} \mathbf{s} = \mathbf{z}^T \mathbf{s}$$

- y is a linear combination of s_i , with weights given by z_i .
- Since sum of two indep r.v. is more gaussian than individual r.v., so $z^T s$ is more gaussian than either of s_i . AND becomes least gaussian when its equal to one of s_i .
- So we could take $\mathbf{w}$ as a vector which maximizes the non-gaussianity of $\mathbf{w}^T \mathbf{x}$.
- Such a $\mathbf{w}$ would correspond to a $\mathbf{z}$ with only one non zero comp. So we get back the s_i .

Measures of Non-Gaussianity

- We need to have a quantitative measure of non-gaussianity for ICA Estimation.
- Kurtosis : gauss=0 (sensitive to outliers) $kurt(y) = E\{y^4\} - 3(E\{y^2\})^2$
- Entropy : gauss=largest
- Neg-entropy : gauss = 0 (difficult to estimate) $H(y) = - \int f(y) \log f(y) dy$
- Approximations

$$J(y) = H(y_{gauss}) - H(y)$$

$$J(y) = \frac{1}{12} E\{y^2\}^2 + \frac{1}{48} kurt(y)^2$$

$$J(y) \approx [E\{G(y)\} - E\{G(v)\}]^2$$

where v is a standard Gaussian random variable and :

$$G(y) = \frac{1}{a} \log \cosh(a.y)$$

$$G(y) = - \exp(- a.u^2 / 2)$$

Computing the rotation step

This is based on the maximisation of an objective function $G(\cdot)$ which contains an approximate non-Gaussianity measure.

$$Obj(\mathbf{W}) = \sum_{t=1}^T G(\mathbf{W}^T \mathbf{x}_t) - \Lambda(\mathbf{W}^T \mathbf{W} - \mathbf{I})$$

$$\frac{\partial Obj}{\partial \mathbf{W}} = \mathbf{X}g(\mathbf{W}^T \mathbf{X})^T - \Lambda \mathbf{W} = \mathbf{0}$$

where $g(\cdot)$ is derivative of $G(\cdot)$,

$\mathbf{W}$ is the rotation transform sought

Λ is Lagrange multiplier to enforce that
is an orthogonal transform i.e. a rotation

Solve by fixed point iterations

The effect of Λ is an orthogonal de-correlation

FastICA

Aapo Hyvarinen (97)

Fixed Point Algorithm

Input: $\mathbf{X}$

Random init of $\mathbf{W}$

Iterate until convergence:

$$\mathbf{S} = \mathbf{W}^T \mathbf{X}$$

$$\mathbf{W} = \mathbf{X}g(\mathbf{S})^T$$

$$\mathbf{W} = \mathbf{W} \sqrt{(\mathbf{W}^T \mathbf{W})^{-1}}$$

Output: $\mathbf{W}, \mathbf{S}$

$\mathbf{W}$

The overall transform then to
take $\mathbf{X}$ back to $\mathbf{S}$ is $(\mathbf{W}^T \mathbf{V})$

There are several $g(\cdot)$ options,
each will work best in special
cases. See FastICA sw / tut
for details.

Application domains of ICA

- Blind source separation (Bell&Sejnowski, Te won Lee, Girolami, Hyvarinen, etc.)
- Image denoising (Hyvarinen)
- Medical signal processing – fMRI, ECG, EEG (Mackeig)
- Modelling of the hippocampus and visual cortex (Lorincz, Hyvarinen)
- Feature extraction, face recognition (Marni Bartlett)
- Compression, redundancy reduction
- Watermarking (D Lowe)
- Clustering (Girolami, Kolenda)
- Time series analysis (Back, Valpola)
- Topic extraction (Kolenda, Bingham, Kaban)
- Scientific Data Mining (Kaban, etc)

Image denoising

Original
image



Noisy
image



Wiener
filtering



ICA
filtering

